OPTIONAL CHALLENGE – PIPELINED MULTIPLIER

1. Design Approach:

Designed an 8-bit pipelined multiplier using Verilog HDL to perform multiplication through partial product generation and addition. Divided each 8-bit input operand into two 4-bit sections and generated four partial products. Implemented pipeline stages to store partial products and calculate the final 16-bit multiplication result.

2. Verification Steps:

- Created a separate testbench.
- Applied reset to initialize the multiplier.
- Provided sample 8-bit input operands.
- Generated four partial products.
- Added shifted partial products to calculate the final result.
- Checked the output valid signal.
- Compared multiplication results with expected values.
- Simulated using ModelSim.

3. Results:

The testbench checks multiplication using three input combinations:

- $10 \times 20 = 200$
- $20 \times 30 = 600$
- $255 \times 255 = 65025$

The output results are compared with expected values. PASS/FAIL messages are displayed for verification.

4. Issues Faced:

Dividing operands into 4-bit sections and generating partial products were challenging. Correct bit shifting and addition of partial products are important to obtain accurate multiplication results. Maintaining valid signal synchronization between pipeline stages is also required.

5. Conclusion:

Successfully designed a pipelined 8-bit multiplier using Verilog HDL to demonstrate partial product generation, multi-stage processing and 16-bit multiplication.